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Task Description:

Currently, information on in vivo biological effects of HZE particles is limited, although the in vivo study is the most relevant approach to assess health risks of exposure to radiation. The overall goal of this study is to characterize the type and the distribution of chromosomal damage and the expression of genomic instability (measured as the delayed appearance of chromosomal damage) in hematopoietic cells (the target cell) following in vivo exposure to 56Fe ions or to 137Cs gamma rays. Two well-characterized inbred strains of mouse known to be radiosensitive (CBA/CaJ) and radioresistant (C57BL/6J) to the induction of chromosomal damage and cancer are used in this study. Therefore, the induction of chromosomal damage and the expression of genomic instability can be evaluated as a function of radiation dose, radiation quality and genetic background of exposed individuals.
We exposed male mice (CBA/CaJ and C57BL/6J) whole body to varying doses of 1 GeV/amu 56Fe ions or 137Cs gamma rays. We collected bone marrow cells and peripheral blood samples from exposed mice at different times up to 3 months post-exposure. Multiple sampling times permit an accurate determination of relative biological effectiveness (RBE) for 56Fe ion-induced chromosomal damage relative to 137Cs gamma rays, as a function of absorbed dose, aberration type, and time after irradiation. The novel technology for genome-wide multicolor fluorescence in situ hybridization of mouse chromosomes (mFISH/SKY) is being used for the detection, analysis, and characterization of chromosome damage in bone marrow cells. In addition to mFISH, the mouse micronucleus (MN) assay in blood erythrocytes is used as an alternative approach for evaluating chromosomal damage in blood samples collected from these mice.

Our in vivo study is aimed to address the following hypotheses: (1) Complex chromosomal exchanges are more prevalent in the BM cells of mice exposed to 56Fe ions than those exposed to similar doses of 137Cs gamma rays. (2) There is a dose-dependent increase in the frequency of chromosome aberrations in bone marrow cells and blood erythrocytes of both strains of mouse exposed to 56Fe ions or 137Cs gamma rays. (3) Genomic instability, expressed as delayed appearance of chromosome aberrations, and clonal expansion of chromosomal exchanges commonly found in mice exposed to 56Fe ions and 137Cs gamma rays are detected in a fraction of surviving cells at later times post-irradiation. (4) Genetic background influences the susceptibility to the induction of in vivo chromosomal damage and the subsequent expression of genomic instability.

The result from this study will provide pivotal information that should lead to a better understanding of dose-response kinetics and of mechanisms by which 56Fe ions induced aberrations can potentially be involved in the induction of genomic instability and cancer. The result also would be important in improving risk assessment of protracted space flight.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Upon completion of our in vivo work proposed in this study, the data will definitely provide pivotal information that should lead to a better understanding of dose-response kinetics and of mechanisms by which 56Fe ions induced chromosome aberrations can be potentially involved in the induction of genomic instability and cancer. This statement is based upon the certainty that chromosome aberrations have been proven to be the most relevant endpoint for detecting carcinogenic activity of suspect agents because of its strong correlation with the induction of cancers. The results also would be important in improving risk assessment of protracted space flight. Both 137Cs gamma rays (the reference radiation) and 56Fe ions are used in this study. The results would therefore facilitate determining whether radiation quality influences the probabilities for increased carcinogenesis from space radiation. Moreover, chromosomal damage is measured at the low-dose level of 56Fe ions (i.e. 10 cGy). Consequently, the results should allow us to make a meaningful comparison for understanding the in vivo RBE of 56Fe ions on the target cells as a function of absorbed dose, aberration type and time. The results also would lead to a better understanding of basic mechanisms of cytogenetic damage potentially involved in cancer induction by these two different sources of radiation. In addition, two strains of mouse are used in our study. Clearly, our results should help determine decisively whether genetic background influences susceptibility to the induction of in vivo chromosomal damage and the subsequent in vivo expression of genomic instability by either 137Cs gamma rays or 56Fe ions.

A. Analysis of Micronucleus in Mouse Blood Erythrocytes
To date, we have completed the evaluation of dose-response relationships for the induction of micronuclei (MN) in blood erythrocytes as a function of radiation quality, radiation dose, time after irradiation and genetic backgrounds of exposed mice. Effects of radiation exposure on the production of red blood cells (erythropoiesis) were also examined. We collected blood samples from exposed male CBA/CaJ and male C57BL/6 mice at different times (i.e. 2, 4, 7 days and 1, 3 months) following whole body exposure to different doses of 1 GeV/amu 56Fe ions (0, 10, 50 and 100 cGy) or 137Cs gamma rays (0, 50, 100 and 300 cGy, at the dose rate of 72 cGy/min using GammaCell40 located at Stony Brook University). At each sacrifice time for each strain of mouse, samples were collected from four groups of five mice in each. We stained blood-smear slides with acridine orange (AO). The analyses of MN were done on coded slides using a Zeiss fluorescent microscope (with FITC filter) equipped with the image analysis system. With the AO staining method, MN appear bright yellow, normochromatic erythrocytes (NCE) or mature red blood cells appear dark green and polychromatic erythrocytes (PCE) or immature red blood cells appear bright red. We examined levels of MN in NCE (MN-NCE) and in PCE (MN-PCE) as well. The ratio of PCE/NCE was determined at each harvest time to monitor the kinetics of effects of 56Fe ions or 137Cs gamma rays on erythropoiesis.

We found a dose-dependent decrease in numbers of PCE at day 2 post-exposure, regardless of radiation quality in both strains of mouse, suggesting a suppression of erythropoiesis from radiation exposure. Subsequently, an increased production of PCE was detected in all exposed mice, reaching control levels or approximately 1.5 times higher by day 7. The increased rate of PCE production was higher in mice exposed to 56Fe ions than those exposed to 137Cs gamma rays. These increases may reflect the compensating or feedback mechanism of hematopoietic cells after injury.

At early time points post-exposure (days 2, 4 and 7), we detected a delay in expression of chromosomal damage in PCE of CBA/CaJ mice exposed to 56Fe ions. This effect seems to be unique in 56Fe ion-exposed CBA/CaJ mice. However, we observed unequivocally dose-dependent increases in levels of MN-PCE in blood samples collected at late times (1 and 3 months) following exposure from CBA/CaJ mice exposed whole body to 56Fe ions or 137Cs gamma rays. This finding indicates the in vivo induction of genomic instability in CBA/CaJ mice by 56Fe ions. The data were described by linear dose–response functions. The RBE values for the in vivo induction of MN-PCE for 56Fe ions were 4.5 and 2.6

Task Progress:	at 1 and 3 months post-irradiation, respectively. In contrast, there was no evidence of in vivo induction of genomic instability in C57BL/6 mice exposed to either 56Fe ions or 137Cs gamma rays at varying doses used in our study. In contrast, we detected a dose-dependent increase in the levels of MN in NCE (MN-NCE) from days 2 to 7 after exposure to either 56Fe ions or 137Cs gamma rays in both strains of mouse. Similarly, the data were described by linear dose–response functions. The RBE values for 56Fe ions for in vivo induction of MN-NCE at days 2, 4 and 7 were 3.3, 3.5 and 4.2 in the CBA/CaJ mouse; and 3.7, 5.3 and 3.9 in the C57BL/6 mouse. There was no difference in MN-NCE levels at 1 or 3 months post-exposure in mice of both strains exposed to either 56Fe ions or 137Cs gamma rays. It is not surprising that there was no increase in MN-NCE at 1 and 3 months because of the limited life span of micronucleated NCE of the mouse (approximately 30 days). These results indicated that acute genotoxic effects of acute exposure to 56Fe ions (measured by the induction of MN-NCE) can be detected up to 7 days post-exposure in both strains of mouse. In summary, our mouse in vivo blood micronucleus data demonstrated that: (1) There is a unique in vivo response to 56Fe ions in exposed CBA/Ca mice at early time post-exposure. (2) 56Fe ions and gamma rays induced in vivo expression of genomic instability in exposed CBA/CaJ mice (but not in C57BL/6 mice), indicating the influence of genetic background in response to in vivo irradiation. This finding showed that a strain-specific difference in radiosensitivity should be considered when selecting strains of mice for evaluating in vivo biological effects of radiation. (3) A single dose of 1 Gy 56Fe ions or 3 Gy 137Cs gamma rays induced a similar level of chromosomal damage and genomic instability in hematopoietic cells of exposed CBA/Ca mice. (4) RBE values changed with time following irradiation. B. Analysis of Metaphase Chromosome Aberrations in Mouse Bone Marrow Cells by means of Multicolor Fluorescence in situ Hybridization Although the mFISH/SKY method is robust and capable of measuring the frequency, the degree of complexity, and the dose response for the production and persistence of simple and complex aberrations over time, it is very time consuming. Due to a very large number of metaphase cells to be scored, the evaluation of genome-wide chromosome aberrations by mFISH is our current major effort and will be continued into year 3 of the study.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
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Abstracts for Journals and Proceedings	Supanpaiboon W, Honikel L, Brooks AL, Rithidech K. "In Vivo Induction of Chromosomal Damage." 16th Annual Space Radiation Health Investigators' Workshop, Port Jefferson, NY, May 2005. 16th Annual Space Radiation Health Investigators' Workshop, May 2005. , May-2005
Articles in Peer-reviewed Journals	Rithidech K, Supanpaiboon W, Honikel L, Brooks AL. "In Vivo Induction of Micronucleus in Mouse Blood Erythrocytes Following Exposure to 56Fe ions or 137Cs gamma rays." To be submitted to Int J Rad Biol, May 2005. , May-2005